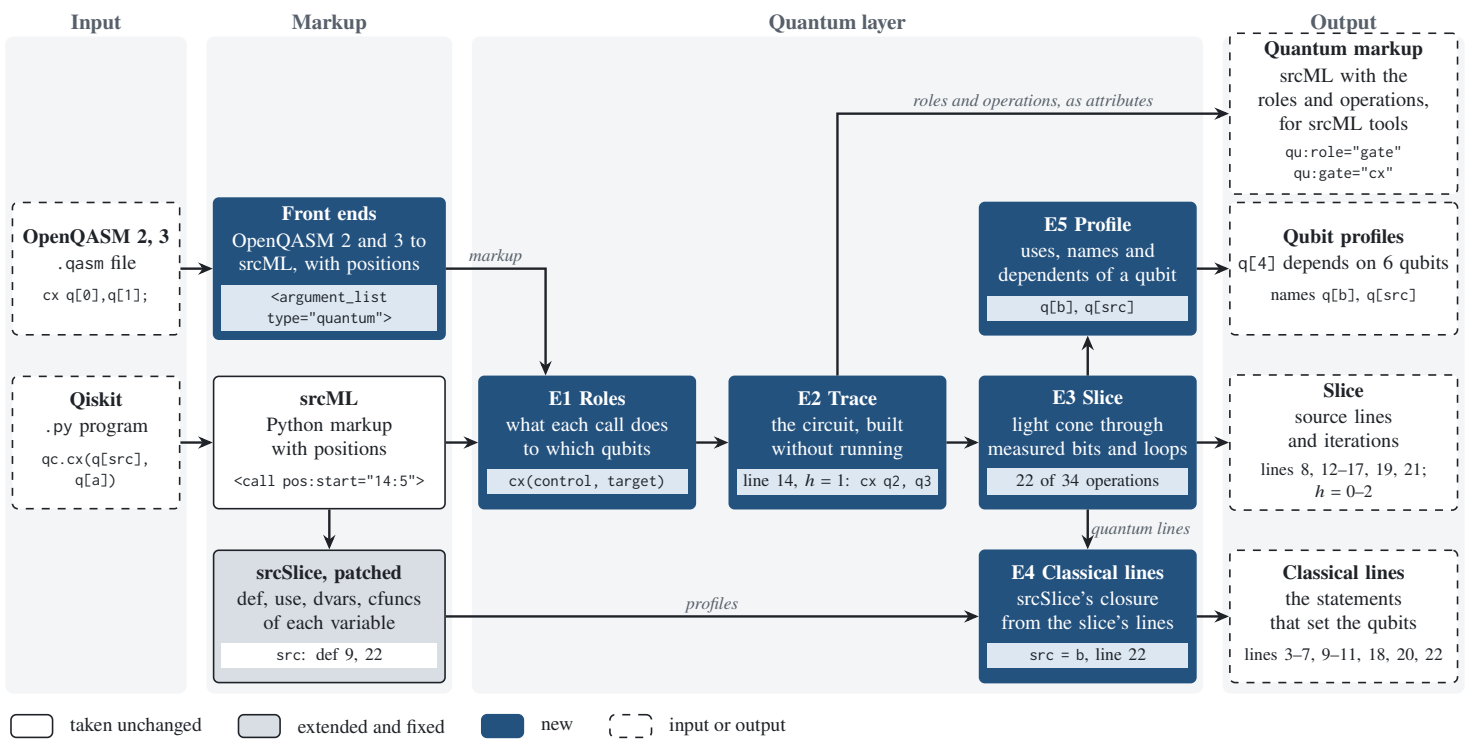

Input
Markup
Quantum layer
Output
OpenQASM 2, 3
.qasm file
cx q[0],q[1];
Qiskit
.py program
qc.cx(q[src], q[a])
Front ends
OpenQASM 2 and 3 to srcML, with positions
<argument_list type="quantum">
srcML
Python markup with positions
<call pos:start="14:5">
srcSlice, patched
def, use, dvars, cfuncs of each variable
src: def 9, 22
markup
E1 Roles
what each call does to which qubits
cx(control, target)
E2 Trace
the circuit, built without running
line 14, h = 1: cx q2, q3
roles and operations, as attributes
E5 Profile
uses, names and dependents of a qubit
q[b], q[src]
E3 Slice
light cone through measured bits and loops
22 of 34 operations
quantum lines
E4 Classical lines
srcSlice's closure from the slice's lines
src = b, line 22
profiles
Quantum markup
srcML with the roles and operations, for srcML tools
qu:role="gate"
qu:gate="cx"
Qubit profiles
q[4] depends on 6 qubits
names q[b], q[src]
Slice
source lines and iterations
lines 8, 12–17, 19, 21; h = 0–2
Classical lines
the statements that set the qubits
lines 3–7, 9–11, 18, 20, 22
taken unchanged
extended and fixed
new
input or output